

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Clockwise in order: Cell (or battery) (1) Voltmeter (1) Ammeter (1) Don't accept ampmeter or voltmeter	3			3		3
		(ii)	To allow a series of readings to be taken or to change the current or the voltage or to change the resistance <u>of the circuit</u>	1			1		1
	(b)	(i)	5 points plotted correctly $\pm$ < one small square (2) ignore no plot plotted at origin 4 points plotted correctly $\pm$ < one small square (1) 3 or fewer points plotted correctly $\pm$ < one small square (0) Straight line drawn through the points and origin $\pm$ < one small square (1)		3		3		3
		(ii)	$R = \frac{V}{I} = \frac{8}{\text{correct reading from graph}}$ (1) Expected answer = 2.5 [Ω] (1)	1	1		2	2	2
		(iii)	Increases (1) Constant (1) Stays constant (1)		3		3		3
		(iv)	(Argument based on $P = V \times I$) so since both V and I increase through the experiment (1) the power $\equiv V \times I$ increases (1) so agree OR By carrying out one calculation using $P = V \times I$ (1) Second calculation (1) so agree To award both marks the conclusion must be present			2	2	2	2
		(v)	Straight line drawn <u>from origin</u> $\pm$ < one small square with bigger gradient than original (1) Double the gradient (i.e. through point (6, 4.8)) (1)		2		2		2
			Question 4 total	5	9	2	16	4	16